VENKATESAN M

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PROFESSIONAL SUMMARY

Results-oriented Full Stack Developer with specialized experience in building Hospital Information Management Systems (HIMS) and Medical Analyzer Integrations. Proficient in Python, Django, and React.js, with a strong technical background in networking protocols (RS-232, TCP/IP). Proven track record of developing scalable modules for Billing, Inventory, and HR management while solving complex hardware-software interface challenges.

TECHNICAL SKILLS

- **Frontend Development:** React.js (Redux, Hooks), HTML5, CSS3, JavaScript (ES6+).
- **Backend Development:** Python, Django Framework, REST APIs.
- **Database:** MySQL, PyMySQL, Database Optimization.
- **Medical Tech & Networking:** LIS Machine Interface, HL7/ASTM Protocols, RS-232, ERBA XL200, HLC, VIDAS.
- **Tools & Core Skills:** Git/GitHub, VS Code, Linux Administration, SDLC, Logical Troubleshooting.

PROFESSIONAL EXPERIENCE

Full Stack Developer **Feb 2025 – Present**
Vesoftometic Solutions

- **HIMS Architecture:** Spearheading the development of core HIMS modules including **HR Management**, **Billing**, Inventory, and Operations using Django and React.js.
- **Form & Data Logic:** Designed dynamic data entry forms with real-time validation logic to minimize human error and ensure data integrity during high-volume patient registration.
- **LIS Integration:** Engineered the LIS Machine Interface to automate data transfer from medical analyzers to the hospital system. Successfully integrated **ERBA XL200**, HLC, VIDAS, Mini VIDAS, H500, H360, and EC90 analyzers.
- **Protocol Troubleshooting:** Resolved complex data parsing issues, including delimiter mismatches in ASTM protocols, ensuring 100% accurate result reporting and analyzer uptime.
- **Connectivity Solutions:** Implemented both RS-232 and LAN-based communication bridges to ensure seamless machine-to-LIS connectivity.
- **Database Management:** Executed complex SQL queries for manual data operations and optimized database schemas to handle high-volume patient records.
- **Deployment & QA:** Led system maintenance and application deployment, performing rigorous testing to validate data synchronization across modules.

KEY PROJECTS

Professional Summary

- Highly motivated and results-oriented professional with strong analytical and problem-solving skills.
- Excellent communication, organizational, and teamwork abilities, committed to delivering quality outcomes.
- Adaptable and proactive, with a focus on continuous learning and professional growth.

Billing & Pharmacy Management Module

- Developed a robust **Billing Module** integrated with pharmacy inventory, featuring automated stock mapping and accurate bill generation.
- Implemented batch-level management to ensure precise pharmacy billing and rectify discrepancies between system entries and physical bills.

HR Management Module

- Designed and deployed a comprehensive **HR Management Module** for employee lifecycle management, including attendance tracking and payroll processing.
- Developed role-based access controls for payroll data, streamlining data flow between frontend interfaces and backend databases while securing sensitive information.

Quality Assurance (QA) Indicators Module

- Developed a monitoring system for blood bank operations to track critical metrics like turnaround time (TAT) and temperature deviations.
- Implemented **Custom Timing Masters** to automate the tracking of time-sensitive blood bank procedures and integrated real-time visual dashboards for audit readiness.

EDUCATION

Bachelor of Computer Science **2021 – 2024**
Bharathidasan University (AVVM Sri Pushpam College, Thanjavur)

CERTIFICATIONS

Python Full Stack Development **Aug 2024 – Nov 2024**
Besant Technologies

- Comprehensive training in full-stack architecture, covering CRUD functionalities, database management, and real-time application deployment.